

STELLA LUCINE MANYAN

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SUMMARY

My research interests are heavily geared towards instrumentation, optical design, and observational astronomy. While I have pursued opportunities in more experimental physical projects, my dream is to one day have a career in astro-instrumentation and operate telescopes!

EDUCATION

Columbia University, New York, NY | GPA: 3.86 Expected Graduation May 2027

- *Majors*: Astrophysics and German Literature & Culture | *Minor*: Mathematics
- *Relevant Coursework*: Observational Astronomy, Modeling the Universe, Optical Systems, Quantum Mechanics I/II, Electricity-Magnetism and Mechanics, Electromagnetic Waves & Optics, Ordinary DE's, Linear Algebra, Astrophysics I/II, Accelerated Multivariable Calculus, Intro to Control Systems (audit)
- *Awards*: Deans list (all semesters), Germanic Department Global Scholarship, Stan Musial Scholarship Recipient, CCE Summer Research Funding Recipient (x2) raising in total \$13000 of research funding
- *Study Abroad*: Freie Universität Berlin International Summer school, B2 German course Summer 2024

TECHNICAL SKILLS

- **Computational**: Ansys Zemax, SolidWorks, SAOImage DS9, Arduino, Python (NumPy, Pyplot, Pandas, Astropy, Statistics), Git, AutoCAD, KiCAD, KLayout for AFM paths, Java, and general astronomical data reduction on Python.
- **Lab Tools**: Atomic Force Microscopy (Oxford, Bruker, and Nanosurf systems), Raman Spectroscopy, Fume Hood Training, 2D Material Nanofabrication, Absolute Arm CMM

RESEARCH EXPERIENCE

MIT Kavli Institute for Astrophysics and Space Research, Optics/Infrared Lab May 2026 – August 2026

- Primarily worked as an opto-mechanical intern to design and fully construct a new optics/camera mount for a dedicated robotic telescope at the Palomar observatory
- Designed the optics in Zemax that incorporated both optical and infrared CMOS detectors to allow for dual imaging. Scientifically, these cameras will contribute to the time domain infrared survey WINTER that operates off an additional port on the telescope
- Constructed the design along with several custom mounting solutions using premium SolidWorks and ran simulations to account for gravitational load and balance
- Worked within the preexisting telescope API to submit targets for observing and learned basics of data reduction using pipelines such as PyeIt and Photutils for both spectral reduction and photometry
- Learned about the specifications of near infrared and optical detectors. Specifically, how read out, material specifications, and signal/noise performance work

Pupin Rutherford Observatory, Columbia University January 2026 – Present

- Led a small student group that focused on upgrading the telescope assembly in the Rutherford dome. Installed a ZWO brand field de-rotator and robotic focuser to allow for precision long exposures
- Additionally constructed a simple fiber fed spectrograph to operate out of the Pupin dome. The project consisted of machining a custom open star cluster plug plate, polishing optical fibers, and assembly of a simple optical bench spectrograph. Spectra were calibrated in the lab and hopefully with plug plate positioning revisions will be tested on sky
- These projects consisted of learning the fundamental principles of observational astronomy and granted exposure to ASCOM compatible astronomy software. All on sky observations and testing was done through operating a Celestron EdgeHD 14" telescope

McIver Lab, Columbia/Max Planck Institute for the Structure and Dynamics of Matter

May 2025 – May 2026

- Student researcher who assists in many projects that consist of nanofabrication of devices comprised of van Der Waals materials
- Built various magic angle twisted graphene and hBN hetero structures
- Optimized a Nanosurf brand AFM to image the moiré potential of these devices using Lateral Force Microscopy
- Have crafted many operating procedures and protocols for the lab. Namely a graphene lithography guide, a hBN thickness color guide, and a guide for making polymer stamps for 2D material transfer
- Attended weekly journal sessions that covered topics from Drude model, band structure, reciprocal and real space, thermal transport, and signal processing through Fourier transform

Columbia Space Initiative

2023 – Present

High Altitude Balloons

- Contributed to payload PCB design of high-altitude Pico Balloon project
- I specifically focused on integrating weather sensors on to a board that could handle up-to-date data transmission and GPS tracking. Additionally, I worked on designs to implement a solar cell system onto the board to power it through flight.
- Researched launch methods, designed basic electronics using KiCad, and successfully conducted multiple balloon launches

RASC-AL: Large-Scale Lunar Crater Prospector (2023-2024)

- As a freshman member of the RASC-AL team, I focused on theoretical power calculations and drill design/implementation for a lunar crater prospector. Our team drafted a full proposal and submitted to the 2024 NASA RASC-AL challenge

OBSERVING EXPERIENCE

2.4 McGraw-Hill Telescope, 1.3 Hiltner Telescope, MDM Observatory, Kitt Peak

March 2026

- Have done in person (4 days) and remote overserving at the MDM facility. Notably spent time observing with the NIR camera MIRAGE. Additionally took time series data of ZTF J1539+5027 which is an ~ 6.91 -minute eclipsing binary system. Other Observations were multiwavelength observations of Messier objects.

NASA IRTF, Maunakea observatories, Hawaii

June 2026

- In two sessions I was instructed on how to do remote observing with the SpeX cross disperser spectrograph and imager. Took follow-up spectral data from WINTER captures of NIR transient events

WINTER Instrument, Plane Wave CDK1000 telescope, Palomar Observatory

May – August 2026

- Throughout the summer I was able to learn how to use the API system for the WINTER instrument to send scheduled observations to the robotic 1-meter telescope it is installed on. Additional observations were done at this facility with on sky testing when installing the new SPRING-SUMMER instrument on the other telescope port

PRESENTATIONS

- **Manyan, S.** “Optomechanical Design and Construction of an Optical and Infrared Robotic Camera Port.” Oral & Poster Presentation, Summer MKI Undergraduate Research Forum, August 2026.
 - o Additionally presented at Columbia poster sessions: Astro-fest, Columbia Physics & Astronomy, September 2026, and the Undergraduate Research Symposium, Columbia University, October 2026
- **Manyan, S.** “Optimization of Lateral Force Microscopy for Robust Moiré Pattern Imaging on the Nanosurf FlexAFM.” *Poster Presentation, Undergraduate Research Symposium, Columbia University, October 2025.*

LEADERSHIP/OTHER UNIVERSITY ACTIVITIES

Intercollegiate Athlete, Columbia Sailing Team 2024 – Present

- As a starting member of the CU sailing team, I practice weekly and travel for nationally recognized regattas in the early fall and late spring to compete against other varsity/club intercollegiate teams.
- I am both a skipper and crew for the team. During the season I am consistently travelling every weekend and practicing around 8 hours/week

Resident Advisor, Columbia University 2024 – Present

- As an employee of the university, I work 20 hours a week overseeing 45 residents.
- Tasks include programming, meeting 1:1 with residents, being on call, heavy cross communication between other University departments and residential life, and generally being a trusted individual that residents can turn to at any time for help

WKCR FM Jazz Department Director/Programmer/Sound Engineer 2023 – Present

- As a co-director of the Jazz Department, I oversee all our programmers and scheduled shows, manage press releases, and train interns seeking to looking to get licensed at the radio station
- At WKCR I run a weekly hour-long jazz show on emerging jazz artists that broadcasts to the greater NYC area.
- As a part of the radio, I am trained to operate the sound board as well as engineer live shows where groups come in to perform

Columbia Armenian Society Vice President 2025 – Present

- We connect with CU alumni and other collegiate Armenian societies across the country to host cultural and advocacy events in the greater NYC and Columbia community
- Previously the secretary of the Armenian Society where I communicated with our 300+ members and released a monthly newsletter

Orens Daily Roast Barista 2023 – Present

- In addition to my studies, I work around 8-15 hours a week at a local cafe to help pay for my tuition
- I am entrusted to perform all roles in the café as an opener, closer, shift leader, cashier, and barista